

AN EXPERIMENTAL INVESTIGATION OF INCOMPRESSIBLE FLOW IN CONICAL DIFFUSERS*

A. T. McDONALD† and R. W. FOX‡

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Summary—Experiments have been performed in clear-plastic diffusers, using water flow discharging into a plenum tank, to obtain performance and flow-régime information for a wide range of conical diffuser geometries.

A line of first appreciable stall has been located with sufficient precision that it is useful in conical diffuser design. Performance data obtained allow solution of the four common optimum problems encountered in diffuser design.

At present these results can be extrapolated only with caution to diffuser geometries other than those tested.

NOTATION

AR	area ratio
C_{PR}	pressure recovery coefficient, $(P_2 - P_1)/(\frac{1}{2}\rho\bar{U}_1^2)$
C_{PRi}	ideal (frictionless) pressure recovery coefficient, $1 - (1/AR^2)$ (one dimensional)
D	diameter of circular diffuser
$\bar{H}_L$	nondimensional head loss ($\bar{H}_L = C_{PRi} - C_{PR}$)
L	length along diffuser wall
N	length along diffuser centerline
p	static pressure
q	dynamic pressure ($\rho\bar{U}^2/2$)
R	local radius of conical diffuser
Re	Reynolds number at diffuser inlet, $\bar{U}_1 D_1/\nu$
$\bar{U}$	average velocity at a section
W	local width of plane-walled diffuser
x	axial length coordinate
η	diffuser effectiveness as commonly defined, C_{PR}/C_{PRi}
ρ	density of fluid
ϕ	diffuser half angle

Subscripts

c	conical
i	ideal
p	plane-walled
w	wall
1	diffuser entrance plane
2	diffuser exit plane

INTRODUCTION

MANY applications of diffusers are found in fluid systems where a need exists to efficiently decrease the kinetic energy of a flow. Although flows in adverse pressure gradients have been subjected to a great deal of research, at present

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† University of California at Davis, Davis, California.

‡ Purdue University, Lafayette, Indiana.

there is no reliable way of predicting diffuser performance. There exists an acute need for further systematic research programs on the behavior of diffuser flows.

Reliable data taken over a wide range of geometries and flow conditions will provide valuable performance information. Such systematic data taken under controlled conditions will be significant in other respects. These data provide a firm basis for conical diffuser design, and they are of use in the development of semi-empirical methods for predicting boundary layer behavior. Additional data are helpful in obtaining a possible correlation among the results of plane-walled, annular and conical diffuser configurations. They provide a fixed reference which will be useful in later investigations of the effects of significant non-geometric variables on diffuser performance.

The purpose of this study was to obtain some of the data needed to satisfy the above goals. To allow direct optical flow visualization, water was used as the flowing medium in diffusers fabricated from clear plastic. Conical diffuser geometry and inlet Reynolds number were varied over a range wide enough to completely specify maps of performance and flow régime.

LITERATURE SURVEY

Comprehensive investigations carried out at Stanford University by Kline and his co-workers¹ on two-dimensional plane-walled diffusers have demonstrated the utility of flow visualization in exploring areas where knowledge of basic flow behavior is incomplete. Although a number of flow régimes had been predicted and/or experimentally determined before, the correlation between flow régime and performance had not been set forth. The results of performance and visual studies for two-dimensional plane-walled diffusers are shown in Fig. 1. The line of first appreciable stall, line $a-a$, of Fig. 1 has been taken from a more recent study of flow régimes in two-dimensional diffusers.² The conditions under which the line has been located are as follows: (a) thin inlet boundary layer compared to passage width, (b) flat velocity profile outside boundary layer at inlet, (c) low Mach number and high Reynolds number, (d) partially blocked exit. The line of maximum performance* (highest C_{PR} at given N/W_1), line $\alpha-\alpha$, is based on the summary of data presented in Ref. 1. The reader is cautioned that the inlet boundary layer does have a significant effect on diffuser losses and performance.³

As shown in Fig. 1, the line of maximum pressure recovery at constant N/W_1 is always located above the line of first appreciable stall at constant N/W_1 . This indicates that there is some stall present in diffusers with maximum performance characteristics.

The voluminous literature on flow in axisymmetric diverging passages has been reviewed.⁴ Although a great deal of data on flows in diverging passages has been amassed in the last 110 years, very little of it has been systematic.

The first known performance prediction for conical diffusers was a map drawn by Eiffel,⁵ which is reproduced in Fig. 2. The method by which the performance curves were obtained was not completely described, but it was suggested⁶ that the results were obtained analytically by accounting separately for the flow losses due to friction and diffuser shape. The original work is unfortunately not available to clarify this point.

Eiffel suggested that the best-compromise diffuser for use in a wind tunnel was one with an area ratio of 9.0 and an included angle of 7 deg. The line of maximum performance, line $\alpha-\alpha$, suggested by the investigations of straight plane-walled diffusers by Kline *et al.*¹ has been added to the Eiffel curves in Fig. 2. The significance of this comparison will be discussed in a later section.

* To allow direct comparison with the results of this study, the centerline length N has been used rather than the wall length L . For moderate diffuser angles N and L are essentially equal.

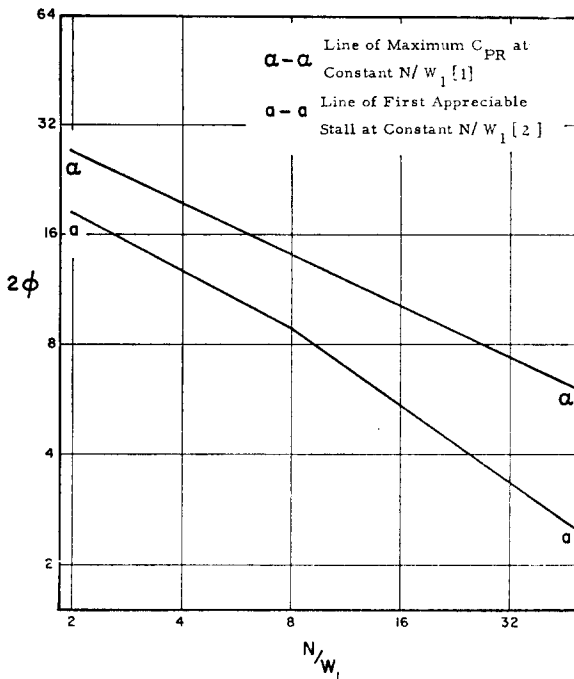


FIG. 1. Lines of maximum C_{PR} and first appreciable stall (at constant N/W_1) for plane-walled diffusers.

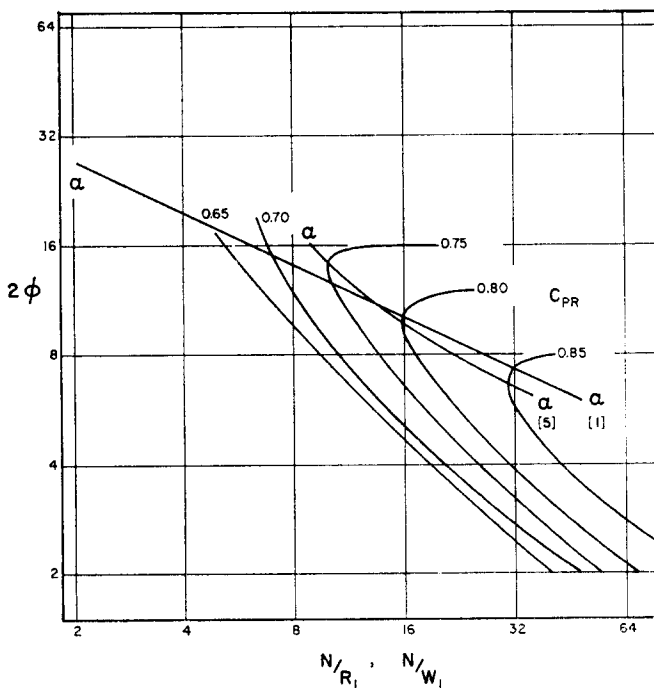


FIG. 2. Performance map of Eiffel (replotted in terms of C_{PR}) compared to line of maximum C_{PR} (at constant N/W_1).

In Gibson's classical investigations^{7,8} conical diffusers were tested over a range of angles from 3 to 180 deg, for area ratios of 2.25, 4.0 and 9.0. These diffusers were tested in a system with a long downstream pipe section for flows with relatively thick inlet boundary layers. Fig. 3 has been prepared from Gibson's original data. Although enough data were not given to complete a performance map, a tentative line of maximum C_{PR} at constant N/R_1 has been drawn from the information available.

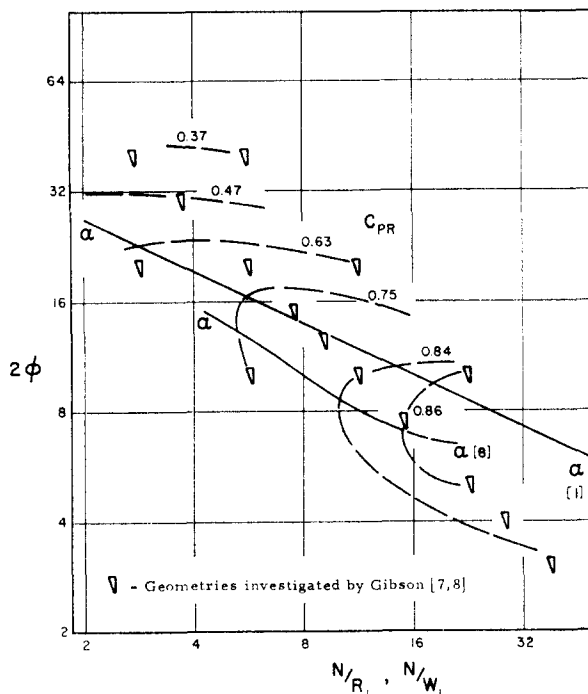


FIG. 3. Map from pressure recovery data of Gibson compared to line of maximum C_{PR} (at constant N/W_1).

Patterson⁹ compared the data of Peters¹⁰ and Gibson. Peters's investigations were made with an air flow system, while Gibson used water, but the exit conditions were similar for both investigations. For similar area ratios, the data showed excellent agreement.

The performance map given by Rippl¹¹ for slim conical diffusers has been replotted in Fig. 4. Rippl tested a number of diffusers, but these included only three different values of divergence angle. The data were obtained for relatively thin inlet boundary layers; the flow discharged from the diffuser as a free jet. The line of maximum performance from the two-dimensional investigations (line $\alpha-\alpha$) is also included in Fig. 4.

Finally, a partial performance map was drawn from the data of Cockrell and Markland¹² and is presented in Fig. 5. Unfortunately, the tests made by Cockrell and Markland included no diffuser angles under 8 deg, and maximum values for C_{PR} could not be established from the curves; data were presented for a range of inlet boundary layer thicknesses with the flow discharging as a free jet from the diffuser exit. The shapes of the curves in Fig. 5 indicate that the line of optimum performance will fall considerably below that from the plane-walled tests. Among the numerous other experimental contributions to the conical diffuser literature, there were not enough systematic data to allow performance maps to be established.⁴

A previous survey and analysis of prediction methods for turbulent boundary layers¹³ has shown that none of these methods gives completely satisfactory results in predicting the location of the line of maximum recovery at a given N/W_1 for plane-walled diffusers. The tentative conclusion, on the basis of limited data, was that application to conical

diffusers also was not successful; this is currently under investigation in the School of Mechanical Engineering at Purdue University.

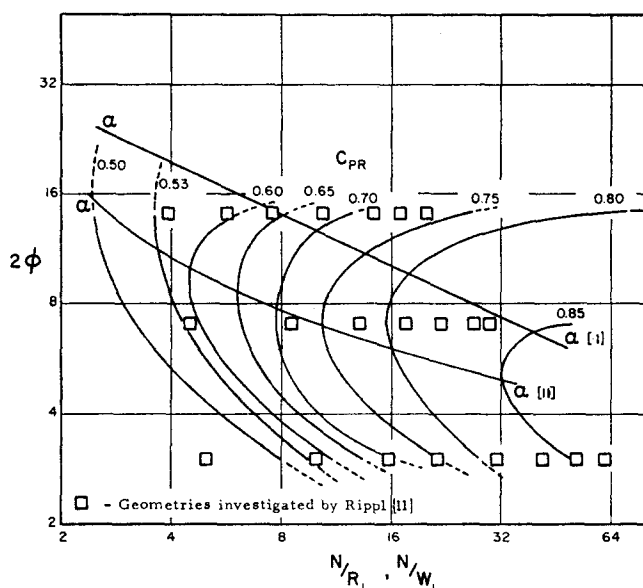


FIG. 4. Performance map of Rippl compared to line of maximum C_{PR} (at constant N/W_1).

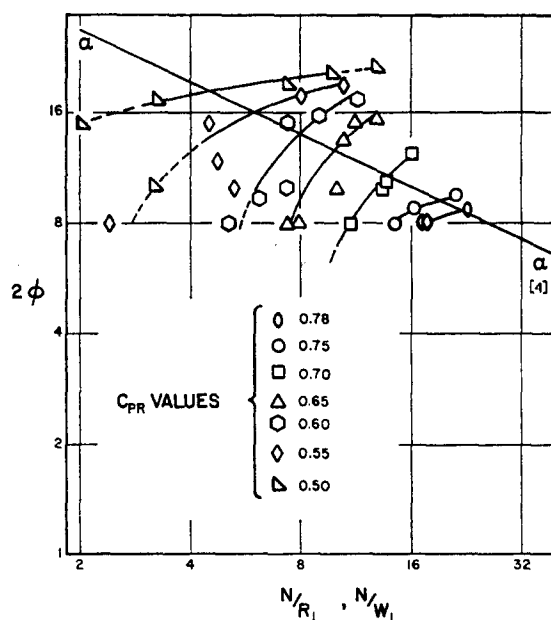


FIG. 5. Map from performance data of Cockrell and Markland compared to line of maximum C_{PR} (at constant N/W_1).

The literature review has emphasized that problems in comparing data from various sources are still present. Several auxiliary variables such as turbulence intensity, Mach number, inlet boundary layer thickness and downstream flow conditions exert their

influence on diffuser performance. Additional difficulties are caused by the dissimilar geometric conditions under which data have been recorded. A summary of experimental data for conical diffusers is given in Ref. 4.

Two pieces of information have recently become available to help in correcting for test conditions. Filippov studied developing turbulent boundary layers in round tubes¹⁴ in 1958. This has made it possible to estimate values of entrance momentum thickness for some previously reported tests. In 1963 Cockrell and Markland¹⁵ correlated the effects of exit lengths on diffuser performance. These results are another step toward correlating additional existing data.

On the basis of the limited conical diffuser performance data available in 1959, it was postulated that performance results for axisymmetric flows could be correlated on the same basis as results from plane-walled diffusers.¹ The suggested correlation was to relate geometric variables as

$$\begin{array}{ccc} \text{Plane walled} & & \text{Conical} \\ 2\phi & \rightarrow & 2\phi \\ N/W_1 & \rightarrow & N/R_1 \end{array}$$

Based on this correlation the resulting curves for maximum performance and first appreciable stall in conical diffusers, when plotted on 2ϕ versus N/R_1 coordinates, would be identical to those shown in Fig. 1 for plane-walled diffusers. However, if the diffuser angle and length ratio are maintained in the correlation, then conical and plane-walled diffusers would have different area ratios. This is shown by the equations

$$\begin{aligned} AR)_p &= 1 + 2 \frac{N}{W_1} \tan \phi \\ AR)_c &= 1 + 2 \frac{N}{R_1} \tan \phi + \left(\frac{N}{R_1} \tan \phi \right)^2 \end{aligned}$$

The result is that for the correlation of variables postulated above all curves for conical diffusers would be shifted when plotted on coordinates involving area ratios.

EXPERIMENTAL DETERMINATION OF CONICAL DIFFUSER PERFORMANCE

A water flow system and plexiglass test sections were constructed with the capability of covering a wide range of diffuser geometries. A photograph of the overall flow system and test section is shown as Fig. 6. The system was designed with sufficient settling tank capacity to give a uniform, swirl-free inlet flow. The approximate design criteria given by Rouse¹⁶ were very helpful. Visualization studies demonstrated that a swirl-free inlet flow could be obtained without additional baffles. The general layout and overall dimensions of the apparatus are shown in the schematic flow diagram, Fig. 7.

In order to study the effects of geometric changes on the diffuser alone, a plenum-type discharge configuration was chosen. The design paper of Kline *et al.*¹ for straight-walled plane diffusers was used in selecting the range of test section geometries. In that paper, it was suggested that conical data correlate with plane-walled data on the basis of 2ϕ and N/R_1 . As shown in Fig. 8, the line of first appreciable stall for plane-walled diffusers was selected as the central line in the map of conical diffuser geometries to be tested.

Unfortunately, the conical geometry did not allow simple variations in geometry. Separate units were built for each diffuser angle tested. Five divergence angles and five lengths were selected on the basis that they provided a minimum of three points along any line on which N/R_1 or AR was constant.

To achieve the required variations in length, each unit was constructed from several demountable sections connected by flanged joints. In addition to serving as a coupling and water seal between diffuser sections, the flanges were so designed that the inside surface of the discharge tank was completely free from projections when coupled to the diffuser flange. Therefore a true free jet exit condition for the diffusers was achieved.

To monitor the static pressure profile along the diffuser, uniformly spaced static pressure taps were drilled into the diffuser wall. The inlet cone to the diffuser was designed,

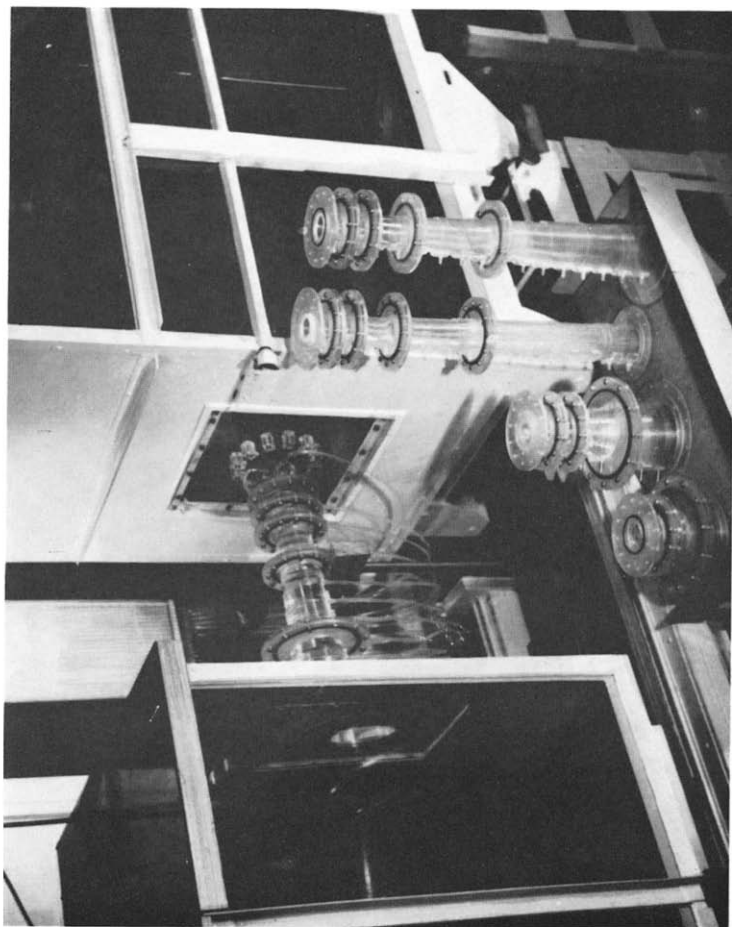


FIG. 6. Overall view of test system.

according to the procedure of Smith and Wang,¹⁷ to give a uniform velocity at the diffuser throat. The losses in the entrance cone were small, allowing the dynamic pressure in the throat section to be measured without inserting a probe. Pressures along the diffuser were measured using a water-filled manometer bank and were recorded with a 35 mm camera.

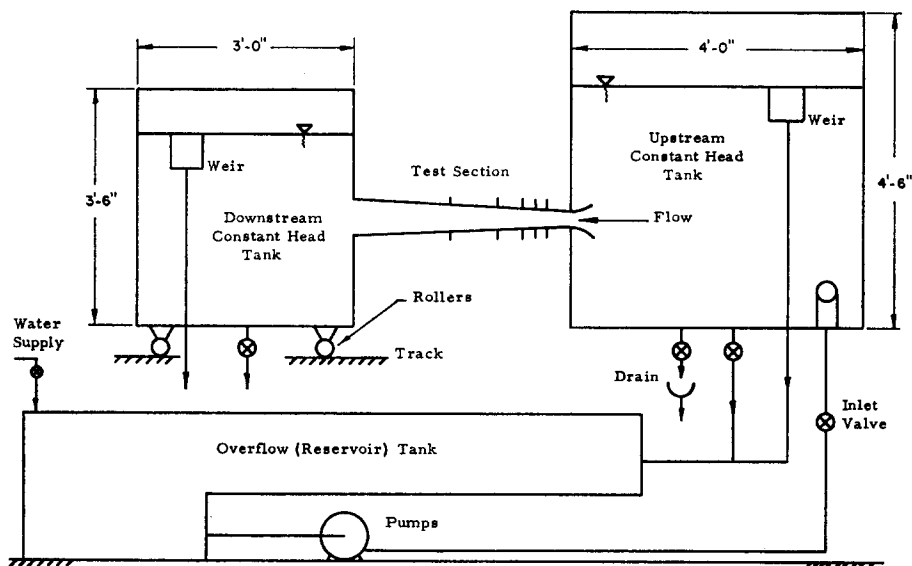


FIG. 7. Water flow system schematic.

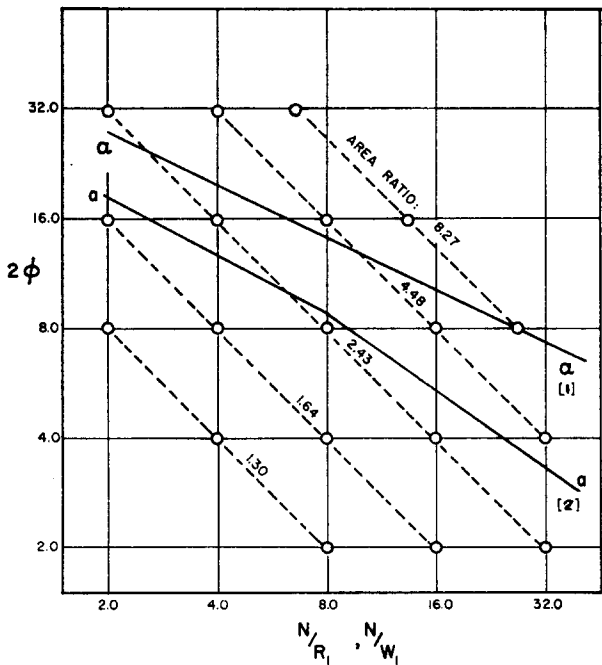


FIG. 8. Conical diffuser geometries selected for study (circles define diffuser geometries investigated).

Preliminary tests run over a range of Reynolds numbers indicated a dependence of diffuser performance on inlet Reynolds number. Fig. 9 is typical of the results obtained when pressure recovery is plotted versus inlet Reynolds number for a given diffuser

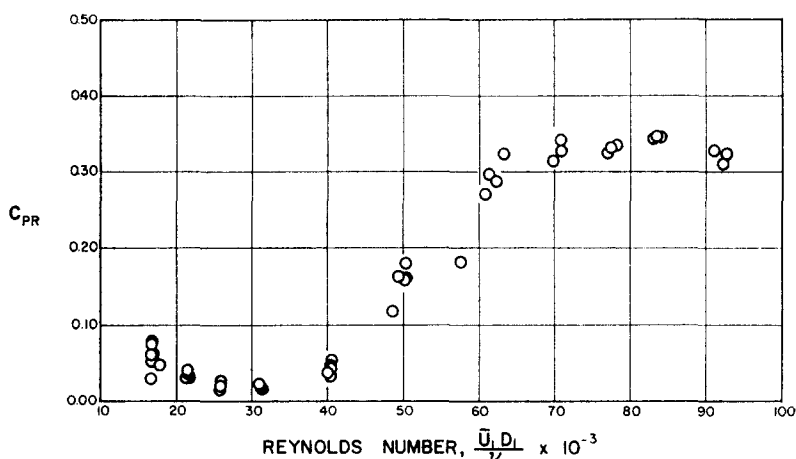


FIG. 9. Effect of low Reynolds numbers on conical diffuser performance ($2\phi = 31.2$, $N/R_1 = 4.0$).

geometry. It can be seen that there is a significant low Reynolds number effect on performance, but that pressure recovery remains constant for the higher Reynolds number flows. This drop in performance with decreasing Reynolds number below a critical value is attributed to a laminar separation immediately downstream of the diffuser throat. For the higher Reynolds number flows the flow is completely turbulent at the diffuser inlet. No variation in performance as a function of Reynolds number was found for any of the tested geometries with inlet Reynolds numbers greater than 75,000.

FLOW VISUALIZATION RESULTS

The only method of flow visualization used in this study was the injection of water base dye. Dye was injected into the boundary layer through the static pressure taps along the diffuser: it was injected with a hypodermic probe into other regions of the flow. The latter method of injection was useful in checking for the possibility of swirl and for determining the extent of any separated regions present.

The behavior of dye traces injected at the wall static taps was essentially analogous to tuft behavior in an air unit. The movements of dye traces were interpreted exactly as tuft movements had been interpreted by Waitman *et al.*³ Classification of tuft movements and symbols to represent them were concisely defined by Waitman *et al.*, and they have been modified only as necessary to apply them to dye traces. The modified definitions have been listed for convenience in Table 1.

The results of all flow-régime observations for $Re > 75,000$ are displayed in Fig. 10 on a map of all geometries tested. The flow condition steadily deteriorated as either diffuser length or diffuser angle was increased. Particularly apparent throughout all the tests was the fact that truly "steady flow" was attainable in the diffuser only for very moderate opening angles and lengths, even though the absolute head difference between the two constant head tanks was closely controlled at all times. For moderate divergence angles, the main flow was steady but the pattern of stall near the wall was not. Only for the most "severe" diffuser configuration tested ($2\phi = 31.2$ deg, $N/R_1 = 6.7$) was the stall pattern almost completely stationary. All other separated flows were to some extent unsteady. The observed flow conditions for each diffuser geometry are described in detail in Ref. 4.

The most noticeable feature of the flow visualization study was that separation phenomena had their greatest effect at the diffuser exit. No significant effect of stalled flow

TABLE 1. DYE TRACE MOVEMENTS AND THEIR INTERPRETATION

Movement and orientation of trace	Type of flow	Symbol
Dye trace held near wall, pointing downstream, occasionally wiggling	Steady flow with occasional disturbance	—
Major part of time dye trace points downstream, wiggling; random flickering (trace quickly points upstream, then downstream again) indicating temporary and local separation	Intermittent transitory stall	I
Dye trace whips continually upstream and downstream indicating rapid and chaotic occurrence and disappearance of separation	Local transitory stall	T
Dye trace whips upstream and downstream major part of time; trace held in upstream position wiggling at random intervals	Local transitory stall with intermittent fixed stall	TIF
Dye trace held upstream major part of time; temporary whipping at random intervals	Local fixed stall with intermittent transitory stall	FIT
Dye trace held upstream, end wiggling	Local fixed stall	F

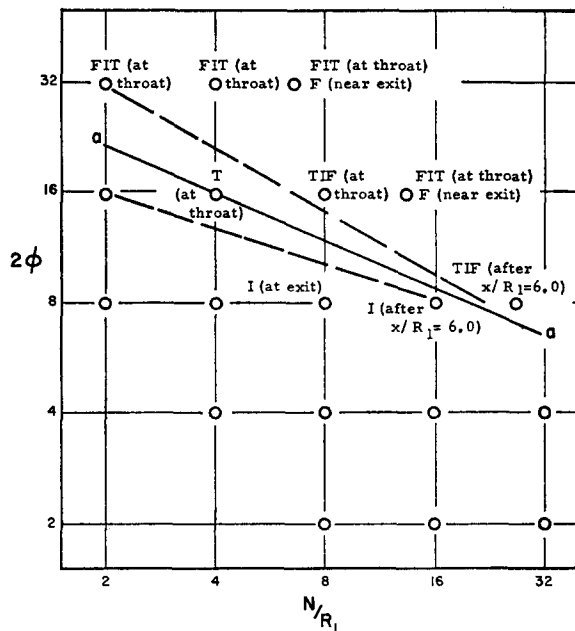


FIG. 10. Flow condition observed for each diffuser geometry investigated (symbols defined in Table 1), showing location of line of first appreciable stall. (Circles define diffuser geometries investigated.)

was noted in upstream regions of the diffusers until the exit conditions had deteriorated considerably past the level of intermittent transitory stall (designated as I in Table 1). Thus the unsteadiness discussed above is attributed to the presence of transitory stall. Even though the presence of intermittent transitory stall (I) at the exit was found not to

influence the flow in the diffuser, this condition was considered to form a lower bound for the placement of a line of "first appreciable stall".

In turn, there seemed to be little question that flow in the 31.2 deg diffusers tested was significantly affected by the amount and kind of stall present. The 15.8 deg, 8.0 length ratio configuration (TIF) was also judged sufficiently affected by stall that the effects were noticeable in the mean flow. Consequently, the upper limit to a possible line of first appreciable stall was located near, but below, the onset of TIF stall (see Fig. 10).

In summary, from Fig. 10 the line of first appreciable stall for conical diffusers is seen to be located fairly well for high values of N/R_1 . For small values of N/R_1 there is some uncertainty in the location of the line as shown by the upper and lower limits on angle. However, the slope of the line is well defined and it is possible to compare it with that for plane-walled diffusers.

CONICAL DIFFUSER PERFORMANCE

The performance data for the diffusers employed in this investigation are presented in Table 2. The flow régime and performance of a conical diffuser are affected by Reynolds number variations only when relatively low-speed flows are being investigated. In practical installations of diffusers, values of Reynolds number are generally at least of the same order as the highest values employed in this study, and in a range where there is no Reynolds number effect.

TABLE 2. PERFORMANCE OF CONICAL DIFFUSERS (WATER UNIT)

2ϕ (deg)	R_1 (in.)	N/R_1	AR	q_1 (psf)	Re_{D_1} ($\times 10^{-5}$)	C_{PR}	η
2.0	1.0	16.0	1.64	61.2	1.25	0.46	0.74
2.0	1.0	32.0	2.43	61.2	1.25	0.64	0.77
4.0	1.0	4.0	1.30	55.6	1.19	0.34	0.83
4.0	1.0	8.0	1.64	61.2	1.25	0.51	0.81
4.0	1.0	16.0	2.43	61.2	1.25	0.66	0.80
4.0	1.0	32.0	4.48	61.2	1.25	0.78	0.82
8.0	1.0	4.0	1.64	61.2	1.25	0.51	0.81
8.0	1.0	8.0	2.43	61.2	1.25	0.64	0.77
8.0	1.0	16.0	4.48	61.2	1.25	0.75	0.79
8.0	1.0	26.8	8.27	61.2	1.25	0.80	0.81
15.8	1.0	2.0	1.64	61.2	1.25	0.42	0.67
15.8	1.0	4.0	2.43	61.2	1.25	0.48	0.58
15.8	1.0	8.0	4.48	61.2	1.25	0.54	0.57
15.8	1.0	13.4	8.27	61.2	1.25	0.59	0.60
31.2	1.0	2.0	2.43	61.2	1.25	0.33	0.40
31.2	1.0	4.0	4.48	61.2	1.25	0.34	0.36
31.2	1.0	6.7	8.27	61.2	1.25	0.35	0.36

In Figs. 11 and 12, contours of constant recovery and constant effectiveness respectively, are plotted on 2ϕ versus N/R_1 coordinates. These curves were obtained from cross-plots of the experimentally measured data. The line of maximum pressure recovery at constant N/R_1 (line $\alpha-\alpha$) for these data has been included in Fig. 11. The location of the line of maximum effectiveness at constant N/R_1 is not clearly defined by the results of Fig. 12. The constant-recovery contours of Fig. 11 have been independently plotted on coordinates of $AR-1$ versus N/R_1 in Fig. 13.

A comparison of Figs. 10 and 11 shows that the line of maximum performance at constant N/R_1 falls below the line of first appreciable stall. (These lines are redrawn in Fig. 14.) The region between the two lines decreases in extent with increasing N/R_1 ; the two lines intersect in the neighborhood of $N/R_1 = 32$. This indicates that conical diffusers designed for maximum performance will be free from any stall which affects the flow significantly.

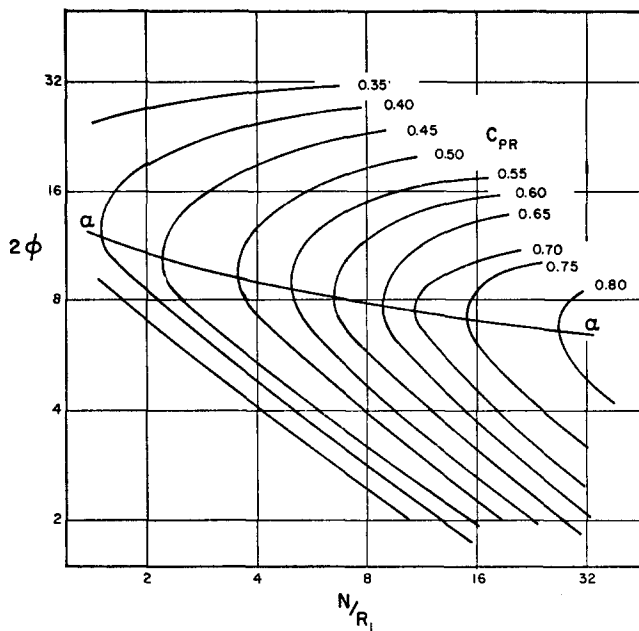


FIG. 11. Pressure recovery contours cross-plotted from conical diffuser data.

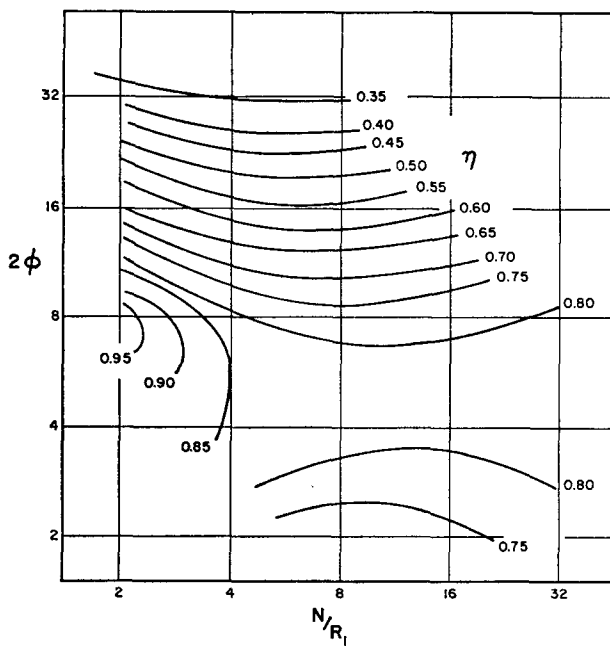


FIG. 12. Effectiveness contours cross-plotted from conical diffuser data.

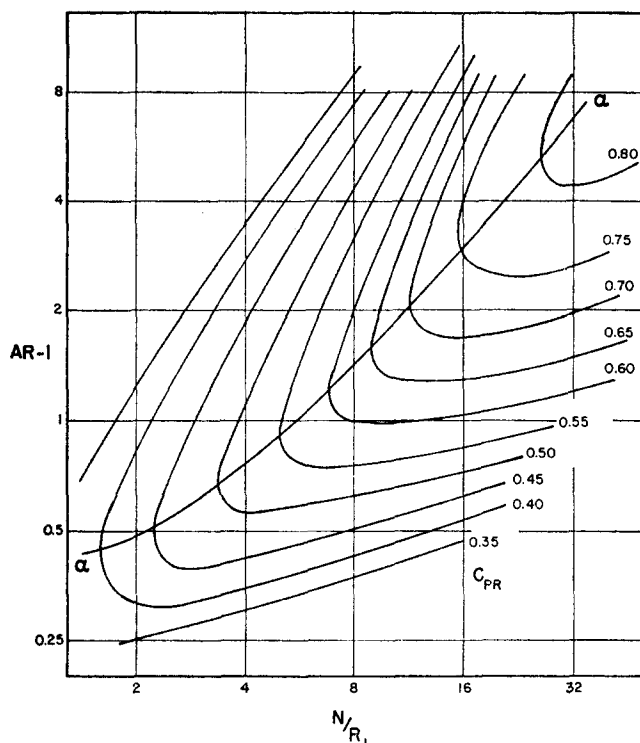


FIG. 13. Pressure recovery contours cross-plotted from conical diffuser data.

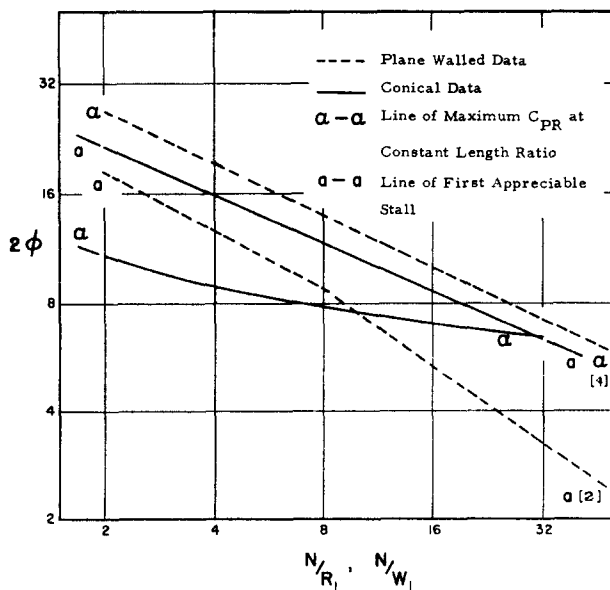


FIG. 14. Lines of first appreciable stall and lines of maximum C_{PR} at constant length-ratio compared for conical and plane-walled diffusers.

Since this finding is contrary to that present in the behavior of two-dimensional plane-walled diffusers¹ it was deemed advisable to check the results with auxiliary data obtained on a separate air unit.* Tests were run for diffusers with $N/R_1 = 4.0$ and 8.0 over a range of angles; the inlet Reynolds number was the same as employed in the water unit; the inlet displacement thickness was 0.013 in. The maximum deviation in the pressure recovery coefficient of the air and water units was 5 per cent. The performance data obtained on the air unit are given in Table 3. The addition of trip wires at the diffuser

TABLE 3. PERFORMANCE OF CONICAL DIFFUSERS (AIR UNIT)

2ϕ (deg)	R_1 (in.)	N/R_1	AR	q_1 (psf)	Re_{D_1} ($\times 10^{-5}$)	C_{PR}	η
4.0	1.0	4.0	1.30	17.6	1.25	0.33	0.81
4.0	1.0	8.0	1.64	17.6	1.25	0.51	0.81
8.0	1.0	4.0	1.64	17.6	1.25	0.49	0.78
8.0	1.0	8.0	2.43	17.6	1.25	0.65	0.78
15.8	1.0	4.0	2.43	17.6	1.25	0.49	0.59
15.8	1.0	8.0	4.48	17.6	1.25	0.57	0.60

inlet had negligible effect on the diffuser performance. Tufts placed on the diffuser walls in the air tests confirmed the flow-régime results of Fig. 10.

DISCUSSION OF RESULTS

Comparison of the line of first appreciable stall obtained from the present conical diffuser experiments (Fig. 10) with the corresponding line for plane-walled diffusers (Fig. 1) shows that the lines of first stall differ in location for the two diffuser configurations. The line of first stall at constant N/R_1 occurs at a larger angle (and area ratio) in a conical diffuser, particularly for higher N/R_1 . Intuitively it would seem that the flow separation would be delayed in a conical diffuser since there are no corners present, and separation in two-dimensional plane-walled diffusers is known to begin in the corners.¹⁸ This type of argument would tend to explain the direction of the shift in the line of first appreciable stall, but it would not necessarily account for the magnitude of the change.

In Fig. 14, lines of maximum pressure recovery at constant length-ratio and lines of first appreciable stall are compared for the conical and two-dimensional geometries. On this plot, not only does the line of optimum performance for the conical units lie below that for plane-walled diffusers, but the shape of the curve is different. This indicates that the relationship $2\phi)_p = 2\phi)_c$, $N/W_1)_p = N/R_1)_c$, as postulated in Ref. 1, is not satisfactory for correlating performance results of conical and two-dimensional plane-walled diffusers. It should be noted that the data used to check the suggested correlation were all obtained for flows with a thin inlet boundary and a free jet discharge from the diffuser exit into a downstream plenum chamber.

In Fig. 15 lines of maximum pressure recovery at constant length-ratio and lines of first appreciable stall are compared further for the conical and two-dimensional plane-walled geometries with the area ratio as a coordinate. The line of maximum pressure recovery, line $\alpha-\alpha$, is steeper in the conical diffuser case. The correlation of the results for the two different geometries is not particularly good. However, the location of the line of maximum pressure recovery at constant N/R_1 of the present study (Fig. 11) compares quite well with the previous results of Rippl (Fig. 4) for conical diffusers.

* Experiments were performed by Mr. Kent Williams, Research Assistant, School of Mechanical Engineering, Purdue University.

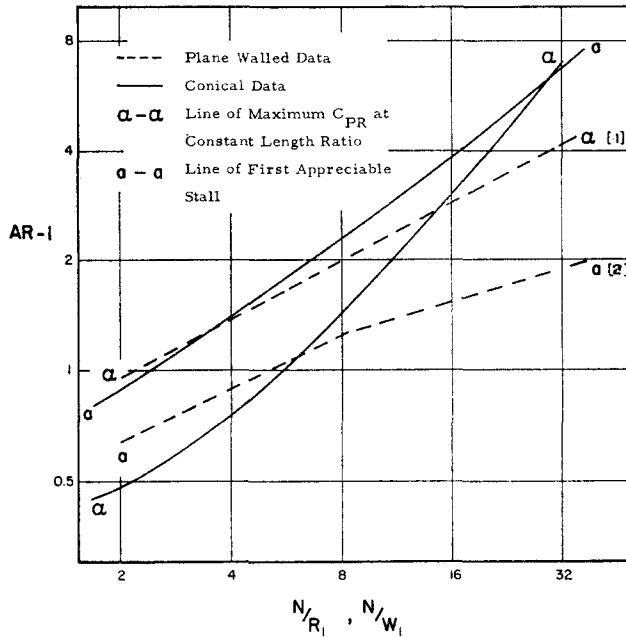


FIG. 15. Lines of first appreciable stall and lines of maximum C_{PR} at constant length-ratio compared for conical and plane-walled diffusers.

DESIGNING FOR OPTIMUM PERFORMANCE

Four common optimum problems often encountered in diffuser design have been discussed previously in Ref. 1. These are:

1. Minimum loss of total pressure for given pressure rise;
2. Maximum recovery for a given area ratio;
3. Maximum recovery for a given length ratio; and
4. Maximum recovery for any possible geometry (best optimum).

For the present study the location of these optima can be determined from Figs. 11 and 13.

Actually, the first two optimum problems listed above have the same solution. Since

$$\bar{H}_L = C_{PR_i} - C_{PR}$$

then minimizing $\bar{H}_L$ for a given value of C_{PR} requires that C_{PR_i} ($C_{PR_i} = 1 - 1/AR^2$) be made as close to C_{PR} as possible. This requires that AR be made as small as possible. The location of the point of minimum area for a given C_{PR} can be determined from Fig. 13.

The point of maximum recovery for a given area ratio (optimum 2 above) clearly lies along the same line as that of the first optimum discussed above.

The line of maximum recovery for a given length-ratio is given as line $\alpha-\alpha$ in Fig. 13.

Finally, optimum problem 4 may be solved by referring to Fig. 11. To obtain best possible recovery, one follows the line of optimum C_{PR} toward the right. For the range of geometries tested, the best recovery is 80 per cent.

CONCLUSIONS

Conical diffusers have been investigated in incompressible flow with low turbulence, thin inlet boundary layer entrance conditions and free jet exit conditions. Original experimental data have been presented and the possibility of correlating conical results with previous plane-walled diffuser work has been explored.

The following are the principal results of the study:

1. For diffuser flows wherein the inlet Reynolds number is greater than 75,000, the pressure recovery, diffuser effectiveness and flow régime are independent of Reynolds number for incompressible flow.
2. The line of first appreciable stall has been located for conical diffusers; there is some latitude in the location of the line.
3. Sufficient data have been presented to allow solution of the four common optimization problems encountered in diffuser design. These solutions are good for the flow conditions investigated, but at the present time extrapolation of the results to diffuser geometries with an exit pipe cannot be accomplished with assurance.
4. The locations of the optima for conical diffusers are not the same as those suggested by data from plane-walled diffusers.
5. Conical diffusers can be designed for maximum recovery; at these conditions the diffusers will be free from flow separation.

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